BIDSIGHT

Run and Demonstration Guide

A direct procedure for running BidSight locally, validating the connection, and presenting the complete three-vendor evaluation.

> **Document purpose**
> Use this guide when setting up the repository or demonstrating the product. It follows the implemented screen order and uses the sample files included with the project.

Document	Value
Document ID	BS-DEMO-001
Version	1.0
Status	Final
Date	12 August 2026
Prepared by	Badar Butt, AI Engineer

BidSight is a procurement decision-support system. Final purchasing authority remains with the authorised user.

1. What the Demonstration Proves

The demonstration shows that BidSight can move from procurement requirements to reviewed quotation evidence, deterministic comparison, and an AI-written recommendation without allowing the model to calculate scores.

Stage	Evidence to show
Evaluation	A purchasing baseline with quantity, budget, currency, delivery target, and requirements.
Quotation intake	Three validated vendor PDFs and clear processing states.
Extraction review	Editable structured fields and explicit missing information.
Scoring	Requirement outcomes and component scores generated in Python.
Recommendation	TechCore selected from verified evidence with cheaper-vendor trade-offs explained.

2. Prerequisites

- Python 3.11 or later and uv.
- Node.js and pnpm.
- A Microsoft Foundry resource endpoint, API key, and deployed model name.
- The repository checked out locally.
- The four PDFs and expected-results.json under sample-data/.

Repository locations used in this guide

```
bidsight/
  backend/      FastAPI application
  frontend/     Next.js application
  sample-data/  Requirements, three quotations, expected results
  docs/         Project documentation
```

3. Configure the Backend

Create an untracked .env at the repository root or inside backend/ using this structure:

```
DATABASE_URL=sqlite:///./bidsight.db
FOUNDRY_ENDPOINT=https://YOUR-RESOURCE.openai.azure.com/openai/v1
FOUNDRY_API_KEY=YOUR_SECRET_KEY
FOUNDRY_MODEL_DEPLOYMENT=gpt-5.6-sol
FOUNDRY_REQUEST_TIMEOUT_SECONDS=90
UPLOAD_DIR=uploads
FRONTEND_URL=http://localhost:3000
MAX_UPLOAD_SIZE_MB=10
```

> **Secret handling**
> Never commit .env or expose FOUNDRY_API_KEY in frontend files, screenshots, logs, or public deployment settings. Rotate a key immediately if it is exposed.

4. Start the Applications

Terminal 1 - backend

```
cd backend
uv sync --group dev
uv run uvicorn app.main:app --reload
```

Expected addresses:

- API root: http://localhost:8000
- Health check: http://localhost:8000/api/health
- Swagger UI: http://localhost:8000/docs

Terminal 2 - frontend

Create frontend/.env.local:

```
NEXT_PUBLIC_API_URL=http://localhost:8000
```

Then start Next.js:

```
cd frontend
pnpm install
pnpm dev
```

Open http://localhost:3000 and confirm the Settings screen reports the API as connected.

Pre-demo health check

Check	Expected result
GET /api/health	HTTP 200 with status ok.
Frontend Settings	Backend connection is available.
Foundry environment	Endpoint, secret key, and deployment name are configured.
Sample files	All four PDFs and expected-results.json are present.
Browser	No CORS or mixed-content errors.

5. Create the Demonstration Evaluation

Field	Value
Title	Computer Lab Laptop Procurement
Category	Laptop
Quantity	25
Maximum budget	4,000,000
Currency	PKR
Required delivery	14 days
Notes	Business laptops for a computer lab; all mandatory requirements must be verified.

Add these mandatory requirements

Requirement	Expected value	Unit	Comparison
Processor	Intel Core i5 or equivalent	-	Match
Minimum RAM	16	GB	At least
Minimum Storage	512	GB	At least
Minimum Warranty	24	months	At least
Delivery	14	days	At most

6. Upload and Process the Quotations

Open the quotation step and upload these files from sample-data/:

Order	File	Vendor
1	techcore-solutions-quotation.pdf	TechCore Solutions
2	digital-systems-quotation.pdf	Digital Systems
3	future-computers-quotation.pdf	Future Computers

BidSight enforces a maximum of three quotations at both UI and API level. Each successful upload should move from UPLOADED to PROCESSING and then READY. If extraction fails, the card shows the backend message and keeps the file available for retry.

What happens during processing

- The API validates the PDF and stores it under the evaluation identifier.
- PyMuPDF extracts and labels readable text by page.
- Foundry returns a strict structured quotation object.
- Pydantic validates the model response before the database is updated.
- The evaluation moves to REVIEW_REQUIRED.

7. Review Extracted Evidence

Open Review and inspect every vendor. Confirm vendor name, product, model, quantity, prices, currency, tax, delivery, warranty, payment terms, support, validity, specifications, source pages, and extraction notes. Correct any field that differs from the source PDF.

> **Required gate**
> Every quotation must be processed and marked reviewed before scoring. BidSight returns a 409 conflict if scoring is requested while any quotation remains unreviewed.

Values to verify

Vendor	Total	Delivery	Warranty	RAM	Storage
TechCore Solutions	PKR 3,625,000	10 days	36 months	16 GB	512 GB SSD
Digital Systems	PKR 3,450,000	18 days	24 months	16 GB	512 GB SSD
Future Computers	PKR 3,300,000	7 days	12 months	8 GB	512 GB SSD

8. Run the Evaluation and Recommendation

Deterministic comparison

Run Evaluation after all quotations are reviewed. The backend calculates requirement outcomes, compliance percentage, price score, technical score, delivery score, warranty score, payment score, support score, overall score, status, and rank.

Expected result

Vendor	Outcome	Reason
TechCore Solutions	COMPLIANT	Meets processor, RAM, storage, delivery, and warranty requirements.
Digital Systems	NON_COMPLIANT	18-day delivery fails the mandatory maximum of 14 days.
Future Computers	NON_COMPLIANT	8 GB RAM and 12-month warranty fail mandatory minimums.

Generate the recommendation

Generate Recommendation only after comparison results are available. The expected recommended vendor is TechCore Solutions. The explanation should state why lower-priced Digital Systems and Future Computers are not eligible.

> **Key demonstration point**
> The cheapest vendor does not win automatically. TechCore is recommended because it is the eligible quotation that satisfies all mandatory evidence. The result comes from source data and deterministic rules, not a hardcoded vendor name.

9. Five-Minute Presentation Script

Time	Action	Message
0:00-0:40	Dashboard and purpose	Explain the manual quotation-comparison problem and the decision-support goal.
0:40-1:20	Create evaluation	Show budget, delivery target, and mandatory requirements.
1:20-2:10	Upload quotations	Show three PDF files, validation, and extraction states.
2:10-3:00	Review	Emphasise that AI data remains editable and must be confirmed.
3:00-4:10	Comparison	Show status, requirement failures, component scores, and price trade-offs.
4:10-5:00	Recommendation	Show TechCore selection and explain that Python owns scoring while Foundry writes the evidence-based summary.

10. Troubleshooting

Symptom	Cause	Resolution
FastAPI command requests fastapi[standard]	Standard CLI extra is missing from the deployed dependency set.	Use fastapi[standard] in pyproject.toml, refresh uv.lock, push, and redeploy.
Frontend shows API unavailable	NEXT_PUBLIC_API_URL is absent, incorrect, or the frontend was not rebuilt.	Set the deployed backend origin and redeploy the frontend.
Browser reports CORS error	FRONTEND_URL does not match the deployed frontend origin.	Set the exact HTTPS frontend origin and redeploy the backend.
Use the Foundry resource endpoint	A project endpoint or unsupported path was supplied.	Use the resource inference endpoint, optionally ending in /openai/v1.
Foundry API key not configured	FOUNDRY_API_KEY is missing.	Add it as an encrypted backend secret and redeploy.
PDF has no readable text	The file is scanned or image-only.	Use a digitally generated PDF; OCR is not implemented.
Scoring is blocked	A quotation is not READY and REVIEWED.	Process, inspect, and confirm every quotation first.
Fourth PDF is rejected	The evaluation already has three quotations.	Remove one quotation or use a different evaluation.

11. Cloud Demonstration Configuration

FastAPI Cloud

- Connect the GitHub repository and set the application directory to backend.
- Use app.main:app as the FastAPI entrypoint.
- Add backend environment variables in the dashboard and mark FOUNDRY_API_KEY as secret.
- Set FRONTEND_URL to the exact deployed Vercel origin.

Vercel

```
NEXT_PUBLIC_API_URL=https://YOUR-BACKEND.fastapicloud.dev
```

Redeploy Vercel after changing NEXT_PUBLIC_API_URL because public Next.js variables are included during the build.

> **Cloud persistence**
> SQLite and local uploads are acceptable for a temporary demonstration. A durable deployment should use PostgreSQL and external object storage because application instances and local files are not a reliable persistence boundary.

12. Reset and Final Checklist

- Delete the demonstration evaluation from Evaluations to remove linked quotations and results.
- Confirm the dashboard, recent activity, and Vendors no longer show the deleted records.
- Keep Foundry secrets out of the repository and rotate exposed keys.
- Before presenting, confirm backend health, frontend connection, sample files, and model deployment.
- Use the video recording as the fallback if a cloud model or network service is unavailable.